

Discussion 3 Solutions

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Problem 1: Sum of First n Odd Numbers

Let $P(n)$ be the statement:

$$1 + 3 + 5 + \cdots + (2n - 1) = n^2$$

Use the Principle of Mathematical Induction to prove that $P(n)$ is true for all natural numbers n by completing the steps below.

1. **Base Case:** Verify that $P(1)$ is true.

For $n = 1$, LHS = 1, RHS = $1^2 = 1 \Rightarrow$ LHS = RHS, so $P(1)$ is true.

2. **Induction Hypothesis:** Assume that $P(k)$ is true for some $k \geq 1$. Write this assumption explicitly.

Assume $P(k)$ is true:

$$1 + 3 + 5 + \cdots + (2k - 1) = k^2$$

3. **Induction Step:** Use the hypothesis to prove that $P(k + 1)$ is true.

Consider $P(k + 1)$:

$$1 + 3 + 5 + \cdots + (2k - 1) + (2k + 1)$$

Using the hypothesis:

$$k^2 + (2k + 1) = k^2 + 2k + 1 = (k + 1)^2$$

So $P(k + 1)$ is true.

4. **Conclusion:** What does this imply about $P(n)$?

By the Principle of Mathematical Induction, $P(n)$ is true for all $n \in \mathbb{N}$.

Problem 2: Factorial vs Exponential Growth

Prove, using the Principle of Mathematical Induction, that for all integers $n \geq 1$,

$$n! \leq n^n$$

Complete the steps below to structure your proof.

1. **Base Case:** Verify that the inequality holds for $n = 1$.

For $n = 1$, we have:

$$1! = 1 \quad \text{and} \quad 1^1 = 1$$

So $1! \leq 1^1$ holds. The base case is valid.

2. **Induction Hypothesis:** Assume that the inequality holds for some $k \geq 1$. Write this assumption explicitly.

Assume that for some integer $k \geq 1$, the inequality holds:

$$k! \leq k^k$$

3. **Induction Step:** Use the hypothesis to prove that the inequality holds for $k + 1$.

We want to prove:

$$(k + 1)! \leq (k + 1)^{k+1}$$

Start with the left-hand side:

$$(k + 1)! = (k + 1) \cdot k!$$

By the induction hypothesis,

$$\leq (k + 1) \cdot k^k$$

Compare the upper bound of the left-hand side to the right-hand side.

$$\begin{aligned} (k + 1) \cdot k^k &\leq (k + 1)^{k+1} \\ &= (k + 1) \cdot (k + 1)^k \end{aligned}$$

This inequality holds because

$$k^k \leq (k + 1)^k$$

which is true for all $k \geq 1$, since increasing the base increases the exponential value.

Therefore:

$$(k+1)! \leq (k+1) \cdot k^k \leq (k+1) \cdot (k+1)^k = (k+1)^{k+1}$$

So the inequality holds for $k+1$.

4. **Conclusion:** What does this imply about the inequality for all $n \geq 1$?

Since the base case holds and the induction step is valid, by the Principle of Mathematical Induction, the inequality $n! \leq n^n$ holds for all integers $n \geq 1$.

Bonus Exercise: Telescoping Series via Induction

Let the sequence $\{a_n\}$ be defined recursively as:

$$\begin{cases} a_1 = \frac{1}{2}, \\ (n+1)(n+2)a_{n+1} = n^2a_n \quad \text{for } n = 1, 2, 3, \dots \end{cases}$$

Prove, using the Principle of Mathematical Induction, that for all natural numbers n ,

$$\sum_{k=1}^n a_k = \sum_{k=1}^n \frac{1}{k^2} - \frac{n}{n+1} \quad (*)$$

Complete the steps below to structure your proof.

1. **Base Case:** Verify that $(*)$ holds for $n = 1$.

For $n = 1$, we have $a_1 = \frac{1}{2}$.

RHS of $(*)$ is:

$$\sum_{k=1}^1 \frac{1}{k^2} - \frac{1}{2} = \frac{1}{1^2} - \frac{1}{2} = 1 - \frac{1}{2} = \frac{1}{2}$$

So LHS = RHS. The base case holds.

2. **Induction Hypothesis:** Assume that $(*)$ holds for some $k \geq 1$. Write this assumption explicitly.

Assume that for some $k \geq 1$,

$$\sum_{i=1}^k a_i = \sum_{i=1}^k \frac{1}{i^2} - \frac{k}{k+1}$$

3. **Induction Step:** Use the hypothesis to prove that $(*)$ holds for $k+1$.

We want to prove:

$$\sum_{i=1}^{k+1} a_i = \sum_{i=1}^{k+1} \frac{1}{i^2} - \frac{k+1}{k+2} \Rightarrow p(k+1)$$

Start from the LHS:

$$\begin{aligned} \sum_{i=1}^{k+1} a_i &= \left(\sum_{i=1}^k a_i \right) + a_{k+1} \\ &= \sum_{i=1}^k \frac{1}{i^2} - \frac{k}{k+1} + a_{k+1} \\ &= \sum_{i=1}^k \frac{1}{i^2} - \frac{k}{k+1} + \frac{k^2}{(k+1)(k+2)} a_k \end{aligned}$$

Substitute for a_k using the recurrence

$$= \sum_{i=1}^k \frac{1}{i^2} - \frac{k}{k+1} + \frac{k^2}{(k+1)(k+2)} \cdot \frac{(k-1)^2}{k(k+1)} a_{k-1}$$

Repeat until reaching a_1

$$\begin{aligned} &= \sum_{i=1}^k \frac{1}{i^2} - \frac{k}{k+1} + \frac{k^2}{(k+1)(k+2)} \cdot \frac{(k-1)^2}{k(k+1)} \cdot \dots \cdot \frac{1^2}{2 \cdot 3} a_1 \\ &= \sum_{i=1}^k \frac{1}{i^2} - \frac{k}{k+1} + \frac{\prod_{i=1}^k i^2}{\prod_{i=1}^{k+1} i^2} \cdot \frac{2}{k+2} \cdot a_1 \end{aligned}$$

Simplify the fraction of two product terms by using telescoping

$$\begin{aligned} &= \sum_{i=1}^k \frac{1}{i^2} - \frac{k}{k+1} + \frac{1}{(k+1)^2} \cdot \frac{2}{k+2} \cdot \frac{1}{2} \\ &= \sum_{i=1}^k \frac{1}{i^2} - \frac{k}{k+1} + \frac{1}{(k+1)^2} - \frac{1}{(k+1)(k+2)} \\ &= \sum_{i=1}^{k+1} \frac{1}{i^2} - \frac{k}{k+1} - \frac{1}{(k+1)(k+2)} \\ &= \sum_{i=1}^{k+1} \frac{1}{i^2} - \frac{k(k+2) + 1}{(k+1)(k+2)} \\ &= \sum_{i=1}^{k+1} \frac{1}{i^2} - \frac{k+1}{k+2} \end{aligned}$$

The last line matches the RHS for $P(k+1)$.

4. **Conclusion:** What does this imply about the identity (*) for all n ?

Since the base case holds and the induction step is valid, by the Principle of Mathematical Induction, the identity (*) holds for all natural numbers n .